



Species identification and strain differentiation of some clinical *Candida* isolates using Restriction Fragment Length Polymorphism analysis

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Article history

Received 08th May 2014

Accepted 04th June 2014

Available online 25th June 2014

Key words:

Candida albicans,
Electrophoresis,
Restriction fragment length
Polymorphism (RFLP) analysis.

ABSTRACT

Candida albicans and non-*albicans* species causing superficial and invasive candidiasis represents an increasing cause of morbidity and mortality in immunocompromised patients. Therefore for understanding their molecular biology, epidemiology and structural population, an accurate fungal characterization is very much essential. The aim of the present study is to identify *Candida* species by using Restriction Fragment Length Polymorphism (RFLP) to discriminate into groups and subgroups according to the similarity in the restriction pattern. Genomic DNA was extracted and purified from 25 clinical isolates of *C. albicans* and non-*albicans*, subjected to restriction digestion with EcoRI and Hind III to detect RFLPs by gel electrophoresis. The RFLP analysis revealed 7 groups of (Gp I-*C. albicans*, Gp II - *C. parapsilosis*, Gp III-*C. tropicalis*, Gp-IV-*C. glabrata*, Gp V-*C. guilliermondii*, Gp VI-*C. krusei* and Gp VII-*C. viswanathi*) fragments ranging from 0.5-10.0Kb and the RFLP profiles were analyzed using UPGMA (unweight pair group method with arithmetic averages cluster algorithm) to form a dendrogram. The UPGMA dendrogram revealed the distribution of similarity values ranging from 0.83 to 1.0 clustering all the 25 isolates into 4 major groups alphabetically as ABCD. The RFLP proved to be an expeditious, responsive, precise and promising method in cataloguing the strains isolated into number of subgroups. The method provides prompt detection and identification of clinically important *Candida* species for the initiation of appropriate antifungal therapy.